

Solar Wind and Space Weather Analytics

An Interactive Big Data Visualization Platform

Project Report for CS661: Big Data Visual Analytics

Department of Computer Science and Engineering, IIT Kanpur

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Ajith Addala

251110006

ajitha25@iitk.ac.in

Aryan Arora

251110016

aryana25@iitk.ac.in

Choudam Jaitej

251110023

jaitejc25@iitk.ac.in

Gaurav Singh Bora

251110031

gauravsb25@iitk.ac.in

Jatin Jangra

251110038

jjangra25@iitk.ac.in

Robin Shah

251110608

rshah25@iitk.ac.in

Tanmay Saurave

251110618

tanmaym25@iitk.ac.in

Source code: https://github.com/cjaitej/BigData_Project

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1. Executive Summary

This project presents an interactive data visualization platform focused on space weather, that is, the state of the near-Earth solar wind and the geomagnetic environment it drives. The platform is built on top of NASA’s OMNIWeb High Resolution (HRO) dataset, a multi-decade, multi-instrument merged record of interplanetary magnetic field and solar wind plasma parameters spanning 1995 to 2026. Designed using a Python data pipeline, a Flask REST API, and a React with D3.js frontend, the tool helps forecasters, researchers, and satellite operators explore decades of measurements, identify geomagnetic storms, understand their seasonal and physical drivers, and reason about the operational risk they pose to spacecraft in different orbits.

The dashboard is organized into five linked, brushable panels arranged around a shared global toolbar for date range, storm catalog, and time-cursor playback. The panels cover raw multi-parameter time series, a phase-space view of the joint solar wind speed and density distribution, a radial view of the seasonal (Russell and McPherron) geomagnetic pattern, a Sankey-style flow of storm escalation pathways, and a physics-informed orbital exposure simulator that animates the compression of Earth’s magnetopause during a storm. Underlying these views is a rigorous acquisition and cleaning pipeline that parses the raw 47-column HRO layout, removes sentinel fill values, derives physically meaningful quantities such as solar wind dynamic pressure, and builds a transparent, rule-based storm catalog. This report outlines the problem context, the theoretical basis, the data handling pipeline, the application architecture, and the design rationale and insights of each dashboard panel.

2. Introduction to the Problem

2.1 Background and Motivation

Space weather refers to the changing conditions in the space environment between the Sun and the Earth, driven primarily by the solar wind, a continuous stream of charged particles and embedded magnetic field flowing outward from the Sun. When this stream carries a strong, southward-oriented magnetic field toward the Earth, it can couple efficiently with the Earth’s own magnetic field and trigger a geomagnetic storm. These storms are not abstract curiosities. They can damage satellites, disturb GPS and radio communications, induce dangerous currents in power grids, and increase radiation exposure for astronauts and high-altitude aircraft.

Over the past several decades, agencies such as NASA and NOAA have collected an enormous volume of measurements from spacecraft positioned in the solar wind upstream of the Earth. These records capture the magnetic field, the plasma speed and density, and a family of geomagnetic indices that summarize the Earth’s response. With careful analysis, this data reveals structure: certain solar wind configurations reliably precede storms, geomagnetic activity is higher around the equinoxes than the solstices, and different satellite orbits experience very different levels of stress during the same event.

The difficulty is that this data is large, multi-dimensional, and physically subtle. A raw table of hourly measurements is nearly impossible to reason about directly, and the relationships that matter, such as the joint distribution of speed and density or the chain of conditions that turns a quiet stream into a storm, are not visible in any single column. This is precisely where interactive visualization becomes valuable. By turning the raw record into linked, brushable charts, the platform

lets a user explore the data at multiple scales and surface the patterns that drive operational risk. Our motivation is to make this rich but forbidding dataset accessible to a research or operations team as a single coherent investigative surface.

2.2 Problem Statement

Space weather data is publicly available at high quality through NASA’s OMNIWeb service, yet accessing it in a usable, analytical form presents a significant barrier. The raw HRO records are delivered as fixed-width text wrapped in HTML, use several different sentinel values to mark missing data, and combine dozens of instrument channels whose physical meaning is not obvious without domain knowledge. As a result, the valuable insight contained in this data often remains out of reach for anyone who is not already an expert in both heliophysics and data engineering.

The core problem this project addresses is the absence of a single, interactive, and physically informed platform that lets a user explore solar wind and geomagnetic storm data across temporal, statistical, categorical, and spatial dimensions at once. Most existing tools present static plots of one parameter at a time, offer no linking between views, and provide no way to move fluidly from a long-term overview to a single storm and then to the operational consequences of that storm for a particular class of satellite.

Specific gaps identified:

- **Fragmented, single-parameter views.** Standard tools plot one channel at a time and force the user to mentally correlate a spike in one plot with a feature in another, rather than linking them directly.
- **No statistical or joint exploration.** There is rarely a way to view the joint distribution of two diagnostic parameters, or to select a cluster in that distribution and see where those points fall in time.
- **Weak temporal-to-operational bridge.** Few tools connect the raw physics of a storm to its consequences, for example the relative exposure of a Low Earth Orbit satellite versus a Geostationary one during the same hour.
- **Opaque storm definitions.** Where storm catalogs exist, the detection rule is often hidden, making it hard to know which events were included or why.

Because of these gaps, the operational and educational potential of a superb public dataset is underused.

2.3 Scope and Objectives

This project is scoped to acquire, clean, and interactively visualize the full span of the OMNIWeb HRO record, from 1995 to 2026, with an emphasis on interactivity, physical grounding, and the ability to move between scales.

Project objectives:

- To build a robust pipeline that downloads, parses, cleans, and enriches the raw HRO data into a single analysis-ready table.
- To detect geomagnetic storm events with a transparent, reproducible rule and classify them by intensity.
- To implement a multi-parameter time-series panel with storm shading and linked brushing.

- To provide a phase-space explorer for the joint solar wind speed and density distribution.
- To visualize the seasonal (Russell and McPherron) geomagnetic pattern in a way that makes its cyclical nature immediately legible.
- To trace storm escalation pathways with a categorical flow diagram.
- To simulate, with a physics-based magnetopause model, the differential exposure of common satellite orbit classes during a storm.

Target users:

- **Researchers and space physicists**, for exploring relationships between solar wind drivers and geomagnetic response.
- **Forecasters and satellite operators**, for assessing which orbital assets are most stressed during a given event and for building an informal library of analog storms.
- **Educators and students**, for learning about the Sun and Earth coupling through hands-on interaction with real data.

2.4 Theoretical Basis: Solar Wind and Geomagnetic Storms

To interpret the dashboard, a few core concepts are needed regarding how the solar wind drives geomagnetic activity and how that activity is measured.

Solar wind and the interplanetary magnetic field. The Sun continuously emits a plasma of protons and electrons known as the solar wind, which carries with it a frozen-in magnetic field called the interplanetary magnetic field (IMF). The key parameters are the bulk flow speed (typically 300 to 800 km/s), the proton number density (a few particles per cubic centimeter), and the strength and orientation of the IMF, measured in nanotesla (nT). The orientation of the north and south component of the IMF, the B_z component in geocentric solar magnetospheric (GSM) coordinates, is the single most important control on whether a storm develops.

Magnetic reconnection and southward B_z . When the IMF points southward (negative B_z), it is anti-parallel to the Earth’s magnetic field at the dayside boundary and can undergo magnetic reconnection. Reconnection opens a pathway for solar wind energy to enter the magnetosphere, driving currents and injecting particles that produce a geomagnetic storm. A strongly negative B_z sustained over several hours is therefore the classic precondition for a storm. Northward B_z , in contrast, tends to leave the magnetosphere quiet. This is why the dashboard treats the sign of B_z as physically meaningful and uses a diverging color scale for it.

Geomagnetic indices: Dst, SYM-H, Kp, and AE. The Earth’s response is summarized by several indices. The Dst (disturbance storm time) index and its high-cadence counterpart SYM-H measure the strength of the ring current that encircles the Earth during a storm; both go strongly negative during storms, and their minimum value is the standard measure of storm intensity. The Kp index is a quasi-logarithmic global activity level from 0 to 9. The AE (auroral electrojet) index tracks the intensity of currents in the auroral zone and responds quickly during the active phase of a storm. These indices give complementary views: SYM-H for the deep ring current, AE for the fast auroral response, and Kp for a coarse global summary.

Storm intensity classification. By convention, storms are graded by their minimum Dst. This project adopts the standard bands: a *moderate* storm has a minimum Dst between -100 and -50 nT, an *intense* storm between -200 and -100 nT, and a *severe* storm at or below -200 nT. Figure 1 illustrates these bands over a synthetic superstorm profile of the kind seen during the October 2003

events.

Dynamic pressure and the magnetopause. The solar wind also exerts a mechanical pressure on the magnetosphere. This dynamic pressure, proportional to density times speed squared, compresses the boundary between the solar wind and the Earth’s magnetic field, called the magnetopause. During strong driving the magnetopause can be pushed inward past the orbit of geostationary satellites, exposing them directly to the harsh solar wind environment. The standoff distance and shape of this boundary can be estimated from an empirical model, which the orbital exposure simulator uses to animate the boundary as conditions change.

The Russell and McPherron seasonal effect. Geomagnetic activity is, on average, higher near the equinoxes (March and September) than near the solstices (June and December). The Russell and McPherron effect explains this through the changing geometry between the IMF and the Earth’s dipole tilt over the course of the year, which modulates how effectively the IMF can present a southward component in the Earth’s frame. The seasonal panel is designed to make this semiannual pattern immediately visible.

2.5 Analyzing Space Weather Data

An OMNIWeb HRO record includes, among other fields, the timestamp, the IMF magnitude and its GSM components, the bulk flow speed, the proton density and temperature, the flow pressure, the electric field, and the geomagnetic indices Kp, AE, and SYM-H. By studying this data one can analyze the frequency and intensity of storms, the joint behavior of solar wind parameters, the seasonal distribution of activity, and the chain of conditions that produces a storm.

Challenges in space weather data analysis.

- (a) **Volume and cadence.** The high-resolution record spans three decades at fine time resolution, so even a single-year request approaches the service’s per-request record limit and must be chunked.
- (b) **Sentinel fill values.** Missing measurements are encoded with a family of distinct fill values that vary by column, such as 999.9, 99999.9, and 9999999, and must be recognized and removed before any statistic is computed.
- (c) **Physical rather than statistical structure.** The relationships that matter are governed by physics, so a naive correlation can be misleading. Southward B_z is a necessary but not sufficient condition for a storm, which is exactly the kind of conditional relationship a flow diagram can reveal but a scatter plot cannot.
- (d) **Bridging scales.** A useful tool must connect a multi-decade overview to a single storm hour and then to the operational consequence of that hour, which requires careful linking between views.

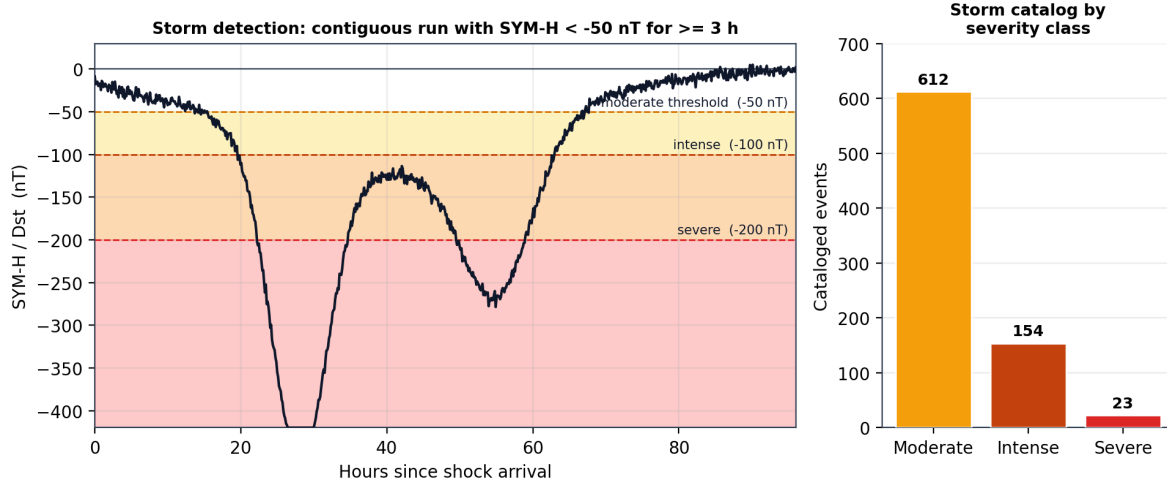


Figure 1: Storm detection and severity classification. Left: a geomagnetic storm is detected as a contiguous run in which SYM-H stays below -50 nT for at least three hours; the shaded bands mark the moderate, intense, and severe intensity classes. Right: an illustrative distribution of cataloged events by severity class.

3. Tasks

This section outlines the specific analytical tasks the dashboard is designed to support:

- Track multiple solar wind and geomagnetic parameters over time and co-locate them with storm intervals.
- Explore the joint statistical structure of the solar wind, especially the speed and density phase space.
- Reveal the seasonal distribution of geomagnetic activity across the calendar year.
- Trace the categorical pathway from a solar wind driver type, through B_z orientation, to a storm outcome.
- Simulate the differential operational risk to satellites in different orbits during a storm.
- Enable user-customizable filtering by date range and by cataloged storm.

3.1 Tasks Addressed by the Visualization Tool

- Multi-parameter, linked time-series tracking with storm shading.
- Bivariate phase-space exploration with marginal distributions and lasso selection.
- Radial, calendar-based visualization of the seasonal geomagnetic pattern.
- Sankey-style categorical flow of storm escalation pathways.
- Physics-informed, animated orbital exposure scoring.
- Shared date range, storm catalog, and playback controls that link every panel.

4. Proposed Methodology of Dedicated Dashboard Panels

Visualization bridges the gap between raw measurements and physical understanding. In the context of space weather, an effective dashboard turns a complex, multi-channel record into linked visuals that support monitoring, research, and operational planning. This project uses a modern web stack, namely a Python data pipeline, a Flask REST API, and a React with D3.js frontend, to build such a dashboard. These tools together enable real-time filtering, custom interactive rendering, and linked views across panels.

Purpose of visualization in space weather analysis. The dashboard is designed around four goals:

- **Pattern discovery**, spotting temporal and statistical structure such as recurring stream patterns or the seasonal activity bulge.
- **Driver identification**, revealing which solar wind configurations precede storms.
- **Operational awareness**, translating physical conditions into per-orbit risk that an operator can act on.
- **Education**, letting non-experts build intuition through direct interaction with real data.

Dedicated panels and their functions. The dashboard provides five linked visualization panels, each with a focused objective: a Time Series Overview, a Phase Space Explorer, a Seasonal Pattern panel, a Threat Escalation Flow, and an Orbital Exposure Simulator. These panels are coordinated by a shared global control bar. Each is detailed in Section 8.

Tools and technologies.

- **React with Vite** for the reactive single-page interface.
- **D3.js** for all custom visualizations, including a hand-built SVG Sankey diagram and a Canvas-based physics animation.
- **Flask with Flask-CORS** for a lightweight REST API over the cleaned dataset.
- **pandas and numpy** for acquisition, cleaning, derivation, and storm detection.

5. Data Sources and Preprocessing

5.1 Overview of the Dataset

The project uses NASA’s OMNIWeb High Resolution (HRO) dataset as its single source. OMNI is a merged, cross-calibrated record that combines measurements from multiple upstream spacecraft into one continuous time series of near-Earth solar wind and geomagnetic conditions, time-shifted to the Earth’s bow shock nose. It is widely regarded as the standard reference dataset for solar wind and magnetosphere coupling studies.

Key features of the dataset:

- **Time period:** 1995 to 2026.
- **Format:** fixed-width numeric records delivered inside an HTML wrapper by the OMNIWeb service.
- **Layout:** a verified 47-column HRO row containing IMF, plasma, and index channels.
- **Scope:** global near-Earth conditions rather than a single location.

The most important columns used by the dashboard are summarized in Table 1.

Column	Description
<code>datetime</code>	Timestamp of the hourly record (UTC).
<code>imf_mag_scalar_nT</code>	Scalar IMF magnitude $ B $ in nT.
<code>by_gsm_nT</code> , <code>bz_gsm_nT</code>	GSM B_y and B_z components; negative B_z is southward and drives storms.
<code>flow_speed_kms</code>	Bulk solar wind speed in km/s.
<code>proton_density_ncc</code>	Proton number density in particles per cubic centimeter.
<code>proton_temp_K</code>	Proton temperature in kelvin.
<code>flow_pressure_nPa</code>	Reported solar wind flow pressure in nPa.
<code>electric_field_mVm</code>	Interplanetary electric field in mV/m.
<code>ae_index_nT</code>	Auroral electrojet index.
<code>sym_h_nT</code>	High-cadence ring-current index (storm marker).
<code>kp</code> , <code>dst_omni</code>	Global activity index (0 to 9) and Dst.

Table 1: Principal HRO columns consumed by the dashboard.

Each hourly record is delivered to the frontend as a compact JSON object. A representative record from the October 2003 superstorm illustrates the shape and the physical range of values the panels work with:

```
{
  datetime: "2003-10-29T06:00:00",
  flow_speed_kms: 721.4, // km/s
  proton_density_ncc: 6.2, // n/cc
  bz_gsm_nT: -32.1, // nT, negative = southward = storm driver
  pdyn_computed_nPa: 7.8, // nPa
  dst_omni: -353.0, // nT, storm if < -50
  kp: 9.0, // 0 to 9
  storm_flag: 1, // 1 during storm periods
  imf_mag_scalar_nT: 46.2,
  ae_index_nT: 2187.0,
  sym_h_nT: -355.0
}
```

The strongly negative B_z , the very low SYM-H, and the maximal Kp in this single record together capture exactly the physics the dashboard is built to surface: a fast, magnetized stream turning sharply southward and driving the ring current to an extreme.

5.2 Data Acquisition Pipeline

Acquisition and preparation are handled by a three-stage Python pipeline built against the OMNIWeb HRO service.

Stage 1: `dataset_extractor.py`. This script downloads every available year from 1995 to 2026 in

one-year chunks. Chunking is necessary because a single request is bounded by the service’s limit of roughly 600,000 records, and a full high-resolution year approaches that bound. Each chunk is saved as a raw text file, and a full variable list is requested so that all IMF, plasma, and index channels are retrieved together.

Stage 2: `parse.html.py`. The OMNIWeb response embeds the numeric data inside an HTML `<pre>` block. This stage streams each raw file, tracks whether it is inside the `<pre>` region, and keeps only the numeric data lines (those beginning with a four-digit year), discarding headers, labels, and blank lines. The result is a set of clean, purely numeric text files ready for parsing.

Stage 3: `pipeline.py`. The final stage, labeled version 4 in the source, parses the verified 47-column layout into named fields. The column indices were confirmed against a known data row so that, for example, index 18 is B_z in GSM, index 24 is the bulk flow speed, and index 44 is SYM-H. It then cleans the data, derives new physical quantities, builds the storm catalog, and writes the analysis-ready table to both CSV and Parquet, along with a `storms.json` catalog.

5.3 Data Cleaning and Validation

Removal of sentinel fill values. The HRO format encodes missing values with a family of distinct fill numbers that differ by column, such as 999.9, 9999.0, 99999.9, and 9999999. The pipeline holds an explicit set of these exact values and replaces every occurrence with NaN before any aggregation, so that a fill value can never be mistaken for a real, extreme measurement.

Physical range checks. Parsed fields are kept only within physically plausible ranges, which catches residual corruption that the fill-value filter alone would miss.

Timestamp construction. The separate year, day-of-year, hour, and minute fields are combined into a single `datetime` column, and the full table is sorted chronologically so that time-based slicing and rolling computations behave correctly.

5.4 Derived Quantities and Feature Engineering

Solar wind dynamic pressure. From the cleaned density and speed, the pipeline computes the dynamic pressure $P_{dyn} = m_p n v^2$. Because the raw fields are stored in convenient observational units, the density n (particles per cubic centimeter) is first converted to SI with $n_{SI} = n \times 10^6 \text{ m}^{-3}$, and the bulk speed v (km/s) with $v_{SI} = v \times 10^3 \text{ m/s}$. Carrying these conversions through, and expressing the result in nanopascals ($1 \text{ Pa} = 10^9 \text{ nPa}$), gives the working form

$$P_{dyn} [\text{nPa}] = m_p n v^2 \times 10^{21} \approx 1.6726 \times 10^{-6} n v^2,$$

where $m_p = 1.6726 \times 10^{-27} \text{ kg}$ is the proton mass, n is in particles per cubic centimeter, and v is in km/s. As a consistency check, the October 2003 values ($n \approx 6.2$, $v \approx 721$) give $P_{dyn} \approx 5 \text{ nPa}$, the correct order of magnitude for strong solar wind driving. Dynamic pressure controls how strongly the solar wind compresses the magnetopause and is a direct input to the orbital exposure simulator.

Normalized channels. Several `*_norm` columns are produced by min and max normalization of B_z , speed, density, AE, dynamic pressure, and IMF magnitude. These normalized values place otherwise incomparable physical quantities on a common 0 to 1 scale so they can be combined into the weighted exposure scores described in Section 8.5.

Solar wind type classification. A categorical `sw_type` field labels each hour by the character of the incoming solar wind stream. This label is the first stage of the Threat Escalation Flow, which traces how each stream type is distributed across B_z orientation and storm outcome.

5.5 Storm Catalog Construction

A geomagnetic storm event is defined as a contiguous hourly run in which SYM-H stays below -50 nT for at least three consecutive hours. For each detected run the pipeline records the start, end, and peak times, the peak SYM-H, the peak Kp, and the peak solar wind speed, and classifies the event as moderate, intense, or severe using the Dst bands defined in Section 2.4. The catalog is written to `storms.json` and is what powers the storm-jump selector and the storm-alignment comparison feature. Because the rule is simple and transparent, the catalog is best understood as a reproducible first-pass screening tool rather than an authoritative event list.

5.6 Specific Dataset Notes

- **Merged provenance.** Because OMNI already merges several upstream spacecraft, no cross-source duplicate resolution is required, unlike catalogs assembled from independent feeds.
- **Default window.** The API defaults to a window around late October 2003, which contains the well-known Halloween storms and provides an immediate, high-contrast example when the dashboard first loads.
- **Dual output formats.** The cleaned table is written as both CSV, for the Flask service to read directly, and Parquet, for efficient re-analysis.

6. Application Architecture

6.1 Overview

The application is a full-stack web platform with a clean separation between a Python preprocessing stage, a Flask service layer, and a React with D3.js presentation layer. Figure 2 shows the end-to-end flow, from the OMNIWeb service through the three-stage pipeline to the cleaned table, then through the REST API into the linked frontend panels.

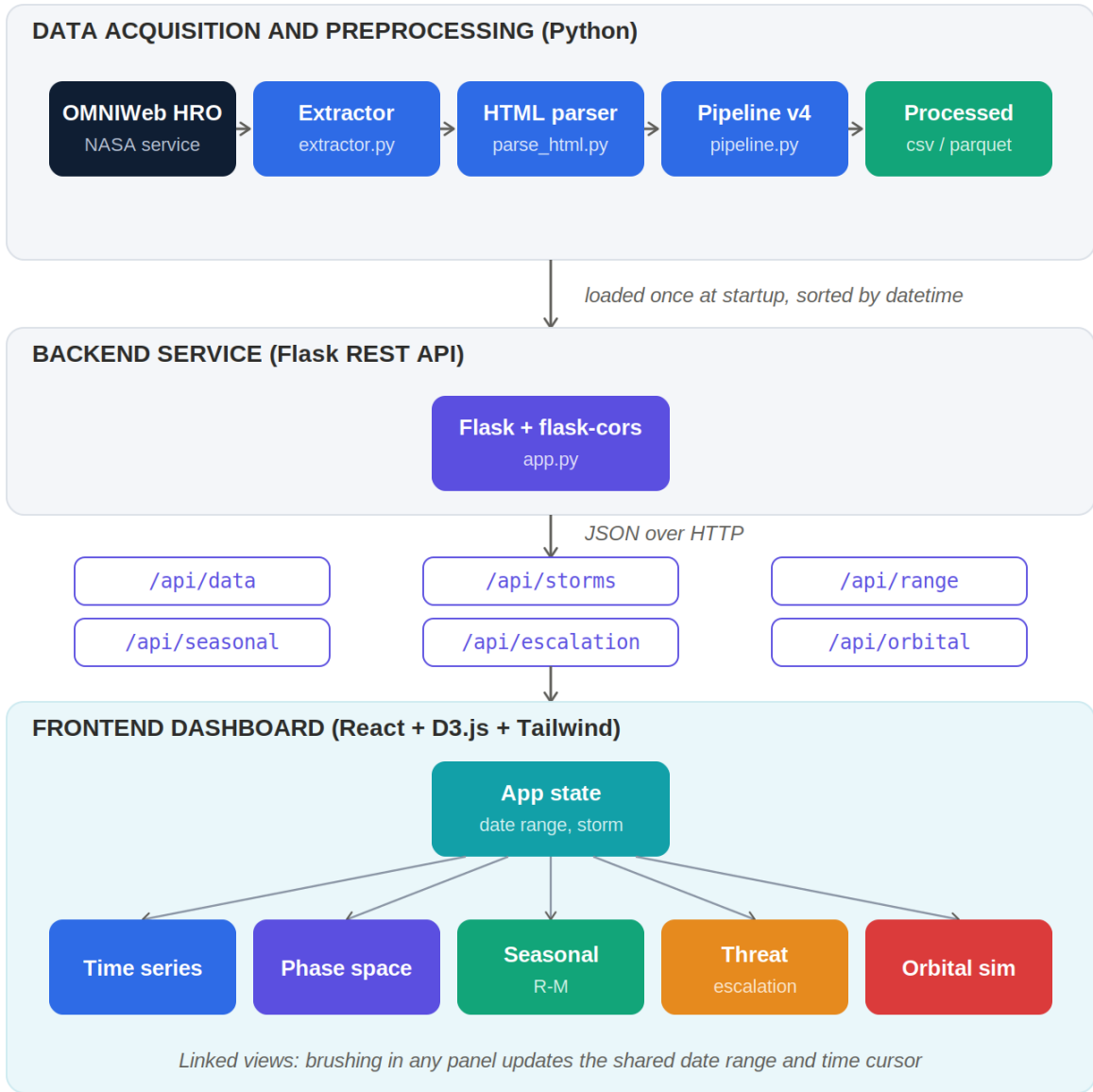


Figure 2: End-to-end system architecture. Raw OMNIWeb HRO data is downloaded, parsed, and cleaned by a three-stage Python pipeline into a single processed table. A Flask REST API serves date-range slices and derived summaries, which a React with D3.js frontend renders as five linked, brushable panels driven by a shared global toolbar.

6.2 Framework and Backend

The backend is a lightweight Flask application with Flask-CORS enabled. At startup it loads the processed CSV once with pandas, parses the `datetime` column, and sorts by time, so that subsequent requests are served by fast, in-memory slicing rather than repeated disk reads. A fixed list of served columns keeps the API payload compact and predictable, and NaN values are converted

to JSON `null` so the frontend can handle gaps cleanly. This design keeps the service simple and stateless while remaining responsive for interactive use.

6.3 REST API Design

The frontend never parses raw data. Instead it requests exactly the slice or summary it needs through a small set of endpoints, listed in Table 2. This separation means the browser holds only the current date range in memory and the heavy lifting stays on the server.

Endpoint	Description
GET /api/data?start=...&end=...	Hourly OMNI records for a date range.
GET /api/storms	Detected storm intervals with min Dst and max Kp.
GET /api/orbital/storms	Full storm catalog with id, intensity, and peak stats.
GET /api/seasonal?start=...&end=...	Monthly mean Kp, AE, and electric field.
GET /api/escalation_flow?start=...&end=...	Hourly driver, B_z , and storm classification counts.
GET /api/range	Full available date range in the dataset.

Table 2: Principal REST API endpoints.

6.4 Frontend Components and Linked Views

The frontend is a React single-page application built with Vite and styled with Tailwind CSS. Each visualization lives in its own component under `src/components/` and receives its data as a prop rather than fetching it independently, which guarantees that every panel is always looking at the same date range. The custom drawing inside each component is done with D3.js against an SVG or Canvas element.

The defining architectural choice is linked views. A brush or lasso in one panel updates a shared application state, which flows back down to every other panel. Selecting a time window in the Time Series panel sets the global date range; lassoing a cluster in the Phase Space Explorer marks the corresponding times in the Time Series panel; and stepping the shared time cursor advances the Orbital Exposure Simulator. The result behaves as one coherent investigative surface rather than five disconnected charts.

6.5 Design Principles

Three principles guided the architecture and are worth stating explicitly, since they explain many of the smaller decisions throughout the application.

Separation of concerns. Acquisition, cleaning, and derivation happen once, offline, in the Python pipeline; slicing and summarizing happen in the Flask service; and only rendering and interaction

happen in the browser. No physical computation is duplicated across layers, which keeps each layer simple to reason about and to test. If the storm rule or a derived quantity changes, only the pipeline is re-run, and every panel inherits the change without modification.

Single source of interaction truth. Rather than letting each panel keep its own notion of the current date range, selected storm, and time cursor, these live in one shared application state. Every panel reads from it and writes to it through well-defined events. This is what makes linked brushing reliable: a selection is not copied between panels, it updates the one shared value that all panels already observe.

Performance-aware bounds. Because the underlying record is dense, the interface imposes deliberate limits, such as the maximum two-year date window and the minimum two-day window, that keep every panel responsive without sacrificing the ability to inspect a single storm in detail. These bounds are a usability feature rather than a limitation, since they prevent the user from accidentally requesting a query that would stall the interface.

6.6 UI and Styling

The interface uses a dark theme for contrast, with accent colors reserved for meaningful encodings such as storm shading and B_z sign. Controls include date-range pickers with pan and zoom, a storm-catalog dropdown, and a playback control, together with informative tooltips that report precise values on hover without cluttering the display.

7. Overall Application Story

The dashboard is designed as an intuitive journey through the dynamics of the Sun and Earth connection, combining temporal, statistical, categorical, and operational layers of analysis into a single understanding of space weather.

The Time Series Overview introduces the user to how the solar wind and geomagnetic indices evolve over the selected window, with storm intervals shaded directly on the charts so that a spike in any parameter can be co-located with an active storm at a glance. From there, the Phase Space Explorer shifts to the statistical dimension, revealing the joint distribution of speed and density that distinguishes one kind of solar wind stream from another. The Seasonal Pattern panel steps back to the scale of the whole year, making the equinoctial activity bulge legible in a single radial glance.

The Threat Escalation Flow then turns to causation, tracing the pathway from a solar wind driver type, through B_z orientation, to a storm outcome, and showing directly that southward B_z is necessary but not sufficient for a storm. Finally, the Orbital Exposure Simulator translates all of this physics into operational consequence, animating the compression of the magnetopause and scoring the exposure of each orbital shell hour by hour through a storm. Underneath every panel is the cleaned, enriched dataset and the transparent storm catalog, and above every panel is the shared toolbar that keeps them in sync. Together they carry the user from a multi-decade overview down to a single storm hour and out to its effect on a specific satellite orbit.

8. Panel-Specific Analysis and Features

For different aspects of the data, the dashboard is divided into five linked panels arranged around a shared global toolbar. Each panel is described below along with the reasoning behind its chosen chart type, the interactions it supports, and the insight it is designed to surface.

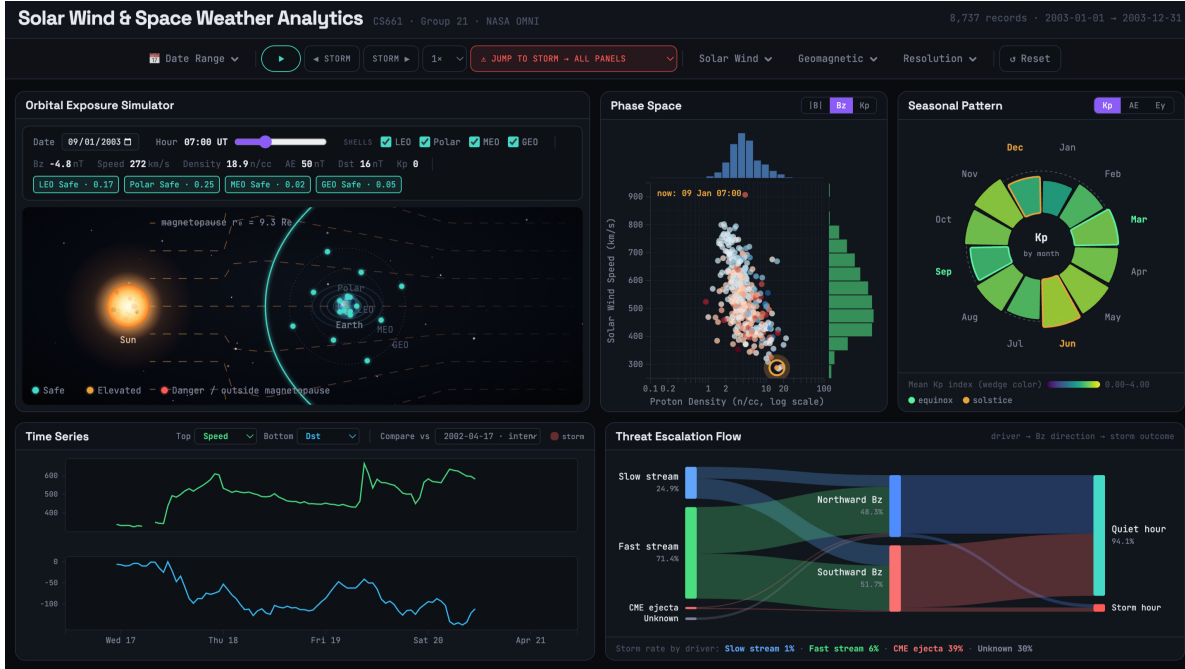


Figure 3: Full dashboard layout: the Orbital Exposure Simulator, Phase Space Explorer, and Seasonal Pattern on the top row; the Time Series Overview and Threat Escalation Flow on the bottom row, all driven by a shared global date range and storm-catalog selector.

8.1 Time Series Overview

8.1.1 Visualization and Controls

The Time Series panel shows two independently selectable, stacked line charts, defaulting to Speed and Dst, drawn from the hourly OMNI record. A dropdown on each row lets the user swap in any of nine parameters: speed, density, B_z , dynamic pressure, Dst, Kp, AE, proton temperature, and IMF magnitude. Storm intervals are shaded in a translucent red wash directly on each chart so that visual co-location between a spike in any parameter and an active storm is immediate.

8.1.2 Interaction

- **Range brushing.** A click and drag draws a selection rectangle that becomes the pending date range for the entire dashboard.
- **Synchronized readout.** A shared crosshair and tooltip report all parameters at the hovered timestamp.

- **Linked selection.** Points lassoed in the Phase Space Explorer are marked here with violet tick lines, a direct brushing-and-linking connection between the two panels.
- **Compare versus storm.** The user can pick a second storm from the catalog and overlay its curve as a dashed line, automatically time-aligned to the main storm’s shock arrival, so that two otherwise unrelated events can be compared on the same relative timeline. This directly supports the question, did this storm evolve like that one.

8.1.3 Backend and Plot Logic

The panel consumes the `/api/data` slice for the current range and renders with D3. Optional transformations such as storm-alignment offsets are computed in the browser. The x-axis adapts to the window: for short windows it labels individual days and hours, and for long windows it places coarser ticks to keep the axis readable.

8.1.4 Insights and Use Cases

The Time Series panel is the entry point for discovering when something happened. It supports detecting storm onset and recovery, comparing the shape of one storm against another through the compare feature, and confirming that a feature seen in another panel corresponds to a specific moment in time.

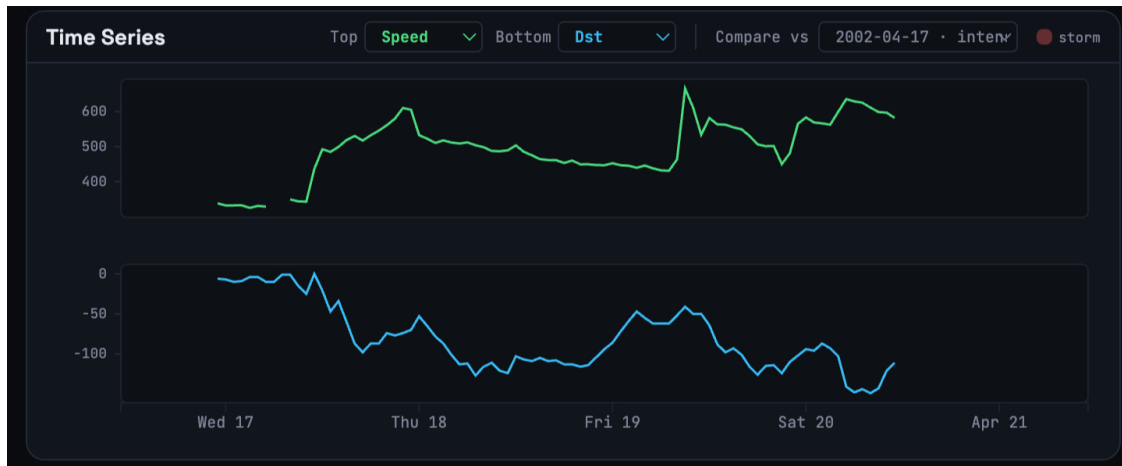


Figure 4: Time Series Overview showing selectable stacked parameter charts with storm shading and a synchronized crosshair readout.

8.2 Phase Space Explorer

8.2.1 Functionality Overview

Solar wind speed and proton density are two of the most diagnostic plasma parameters, and their joint distribution, rather than either in isolation, is what distinguishes solar wind types such as fast coronal-hole streams from slow, dense streamer-belt wind. This panel plots them as a scatter in phase space, with marginal histograms of density on top and speed on the side, so that the univariate distributions remain visible without losing the joint structure.

8.2.2 Interactive Elements

- **Third-channel color.** Each point can be colored by a third channel selected by the user: IMF magnitude or Kp on a sequential scale, or B_z on a diverging scale, since the sign of B_z is physically meaningful.
- **Lasso selection.** The user can lasso a region of points; the selection is linked back into the Time Series panel so the temporal location of a phase-space cluster can be inspected immediately.
- **Selection summary.** Summary statistics for the selection, such as mean B_z and point count, are shown in a small card.

8.2.3 Insights and Use Cases

Because the two solar wind populations occupy different regions of the speed-density plane, the panel lets a user visually separate stream types and then, through the lasso link, ask when a given population occurred and whether it was associated with southward B_z . This makes it a natural tool for characterizing the drivers present in a chosen window.

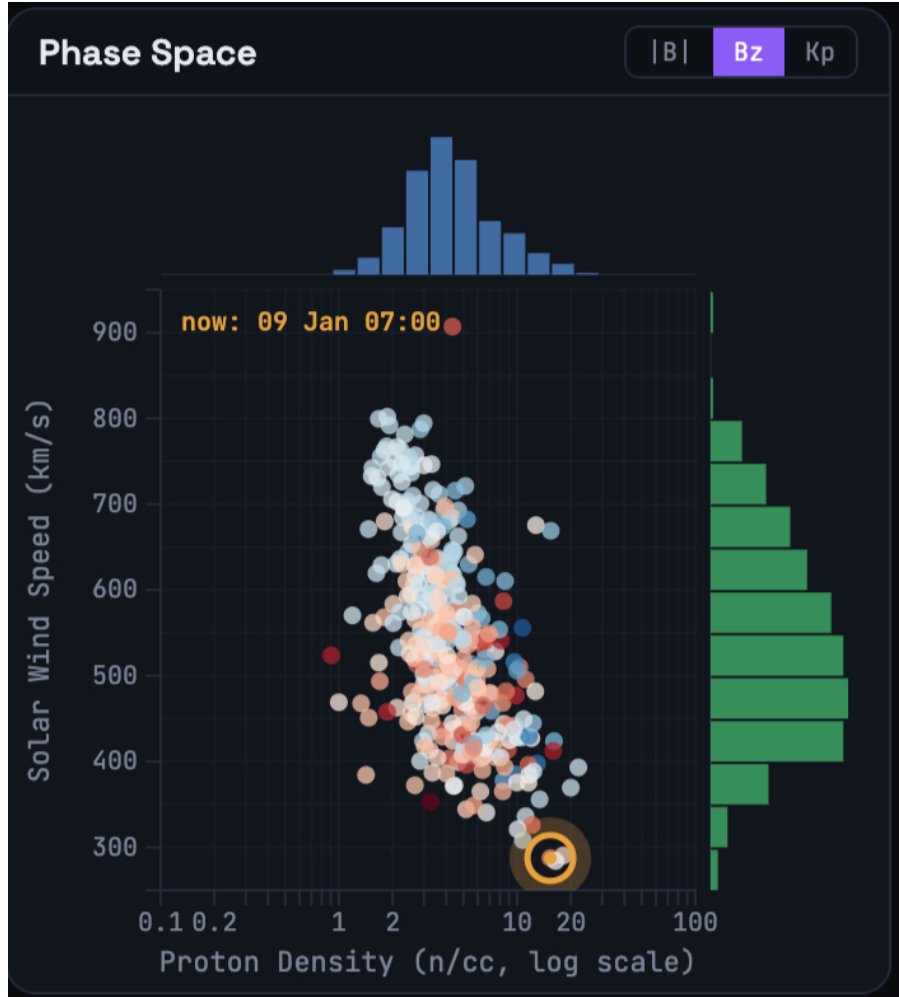


Figure 5: Phase Space Explorer: flow speed versus proton density, colored by a selectable third channel, with marginal histograms and lasso selection.

8.3 Seasonal Pattern: the Russell and McPherron Effect

8.3.1 Functionality Overview

Geomagnetic activity is, on average, higher around the equinoxes than the solstices, the Russell and McPherron effect, caused by the changing geometry between the interplanetary magnetic field and the Earth's dipole tilt over the year. To make this pattern immediately legible, the panel aggregates mean Kp, mean AE index, and mean dawn-to-dusk electric field by calendar month, scoped to the currently selected global date range, and plots the result as a radial bar chart with January at the twelve o'clock position proceeding clockwise. This layout is chosen specifically because it makes a cyclical, calendar-based quantity intuitive in a way a standard bar chart would not.

8.3.2 Interactive Elements

Equinox months (March and September) and solstice months (June and December) are highlighted in contrasting colors so the expected semiannual bulge is immediately visible. Bars are additionally colored on a sequential scale by magnitude, and a legend swatch mirrors the active color-scale domain.

8.3.3 Insights and Use Cases

Because this panel recomputes its aggregation for whatever date range is loaded elsewhere in the dashboard, a user can directly test whether the semiannual pattern holds within a specific solar-cycle phase or a specific set of years, rather than only over the full multi-decade average. This turns a textbook effect into something the user can interrogate against the data.

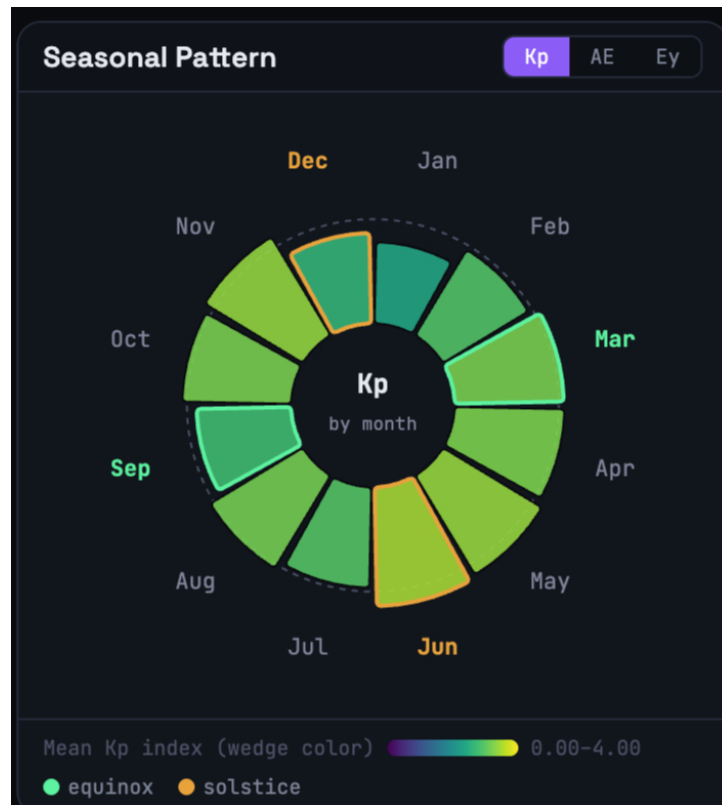


Figure 6: Seasonal Pattern radial chart: mean geomagnetic activity by calendar month, highlighting the equinoctial (Russell and McPherron) effect.

8.4 Threat Escalation Flow

8.4.1 Why This Chart

A storm does not simply happen. It is the outcome of a specific chain of conditions: a type of incoming solar wind stream, whether its embedded magnetic field turns southward, and whether

that combination actually produces a storm hour. We wanted a chart that shows this as a pathway rather than as a single correlation coefficient.

8.4.2 Design and Underlying Logic

The panel is a three-stage Sankey-style diagram, hand-built directly in SVG, that flows from driver type, to B_z direction, to storm outcome. Node heights are proportional to the number of hourly records taking that path, and ribbon widths are proportional to the flow volume between stages, all computed from an hourly classification of the currently loaded date range and served by the `/api/escalation_flow` endpoint.

8.4.3 Insights and Use Cases

The view lets a user immediately see, for example, what fraction of southward B_z hours actually escalate into a storm hour versus remaining quiet, broken down by the type of solar wind stream that carried them. This directly supports the physical intuition that southward B_z is a necessary but not sufficient condition for a storm, and it can be used routinely to audit which driver types are contributing most to storm hours in a given period.

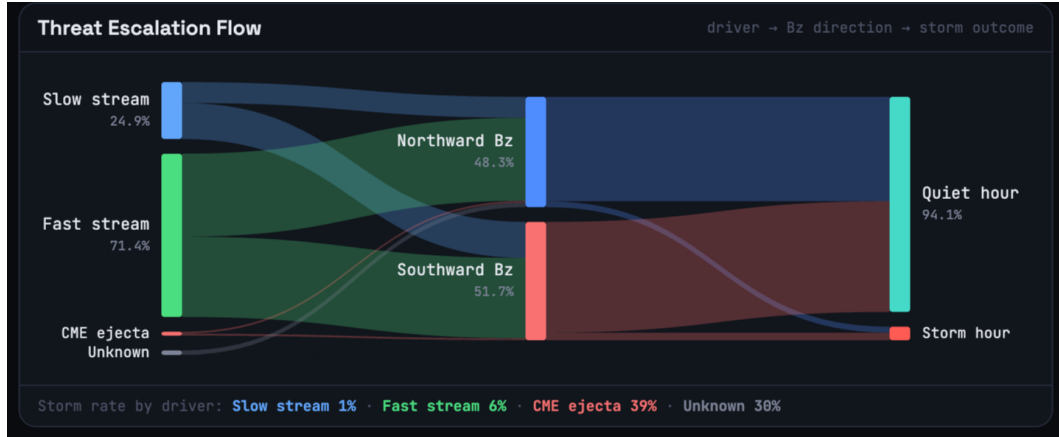


Figure 7: Threat Escalation Flow: a three-stage Sankey diagram tracing solar wind driver type, through B_z orientation, to storm outcome.

8.5 Orbital Exposure Simulator

8.5.1 Core Visualization

Solar wind and geomagnetic conditions do not affect every satellite orbit equally. A spacecraft in Low Earth Orbit (LEO) experiences a very different radiation and particle environment than one in Geostationary Orbit (GEO). To communicate this, the panel presents a Canvas-rendered, Earth-centric scene showing four representative orbital shells at true Earth-radii scale: LEO at about $1.06 R_E$, Polar at about $1.13 R_E$, MEO at about $4.17 R_E$, and GEO at about $6.61 R_E$, together with an animated depiction of the Earth's magnetopause boundary.

8.5.2 Physics Model

The magnetopause standoff shape is computed using the empirical Shue et al. (1998) model,

$$r(\theta) = r_0 \left(\frac{2}{1 + \cos \theta} \right)^\alpha,$$

where the sub-solar standoff distance r_0 and the flaring parameter α are both functions of the instantaneous IMF B_z and dynamic pressure P_{dyn} for the selected hour. As conditions worsen, the modeled magnetopause visibly compresses toward the Earth, and can move inside geostationary distance during a severe storm.

8.5.3 Exposure Scoring

For each orbital shell the panel computes a 0 to 1 exposure score as a weighted combination of the snapshot’s normalized parameters. For GEO, for example, the score weights the magnetopause compression factor most heavily, together with normalized AE and IMF magnitude terms, whereas LEO and Polar scores combine Dst, Kp, density, and B_z terms appropriate to the exposure mechanism dominant at that altitude. Figure 8 shows an illustrative weighting. Each score is thresholded into Safe (below 0.34), Elevated (0.34 to 0.62), or Danger (0.62 and above), shown both as a colored ring around each shell and in a status legend.

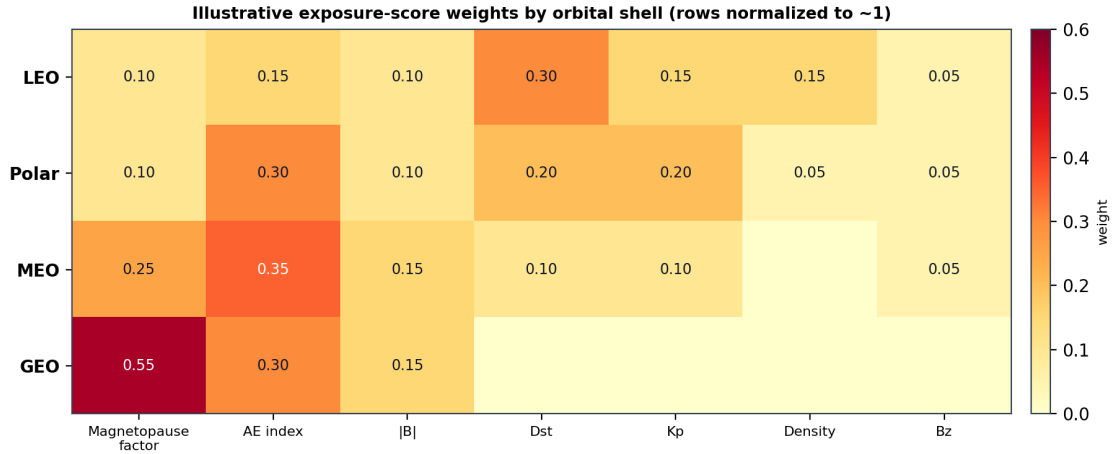


Figure 8: Illustrative exposure-score weighting by orbital shell. Each row combines normalized physical parameters appropriate to the dominant exposure mechanism at that altitude; GEO is weighted most heavily toward magnetopause compression and AE.

8.5.4 Interaction and Use Cases

A date and hour scrubber lets the user step through a specific storm hour by hour, and this same time cursor is shared with the Time Series and Phase Space panels through the global playback control, so the evolving orbital risk picture can be watched frame by frame across an entire storm’s lifecycle. Operators of GEO and MEO assets can use the danger-threshold alerts to prioritize attention during cataloged severe and intense storms.

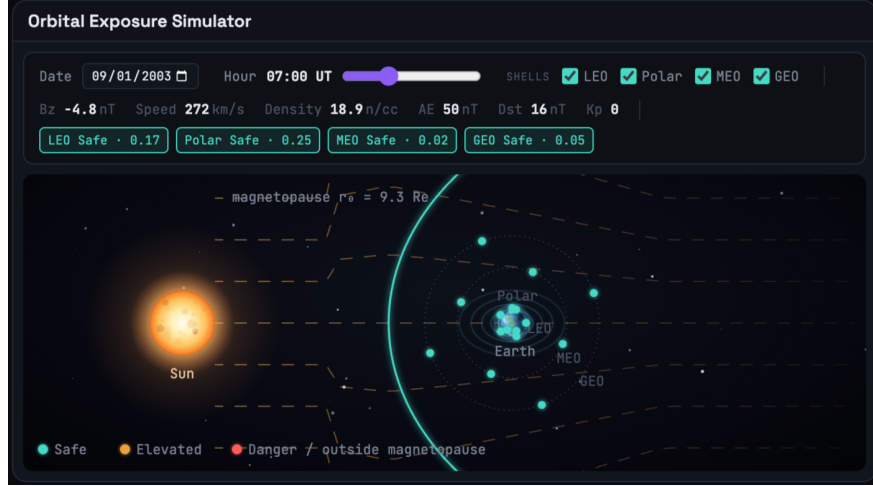


Figure 9: Orbital Exposure Simulator: an Earth-centric view with LEO, Polar, MEO, and GEO shells, a Shue-model magnetopause boundary, and per-shell exposure scoring for the selected date and hour.

8.6 Global Controls: Date Range, Storm Catalog, and Playback

8.6.1 Design Rationale

All five panels share a single global toolbar rather than each maintaining independent filters, so that changing the date range, jumping to a cataloged storm, or stepping the time cursor updates every panel consistently. This is a deliberate choice to keep the dashboard usable as one coherent investigative surface.

8.6.2 Controls

The toolbar exposes a date-range picker with pan and zoom, bounded to a maximum two-year window for performance and a minimum two-day window so single-storm detail views remain meaningful; a storm-catalog dropdown, sorted by severity, that jumps every panel directly to a chosen event; and a play and pause control that sweeps an hourly time cursor through the loaded window. The cursor is visible simultaneously as a moving marker in the Time Series and Phase Space panels and as the driving snapshot time in the Orbital Exposure Simulator.

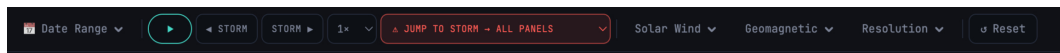


Figure 10: Global menu bar: date-range controls, storm-catalog jump selector, and the shared playback control that drives the time cursor across all linked panels.

9. Challenges and Future Work

9.1 Challenges and Limitations

Several challenges arose during development, some of which constrained the current functionality:

- **Data volume and request limits.** A full high-resolution year approaches the OMNIWeb per-request record limit, which required careful chunked downloading and made the initial acquisition slow.
- **Fragile source format.** Extracting numeric records from the HTML wrapper and mapping the correct 47-column layout was error-prone and required verifying column indices against a known data row.
- **Fill-value handling.** Missing data is encoded with several distinct sentinel values that vary by column, and any that slipped through would masquerade as extreme physical measurements.
- **Rendering performance on long ranges.** Wide date ranges produce dense charts, so the date window is bounded and rendering was tuned to keep interaction responsive.
- **Model simplification.** The exposure score is a transparent weighted heuristic rather than a validated radiation model, and should be read as a relative comparison across orbits rather than an absolute dose estimate.

9.2 Future Enhancements

- **Real-time data integration.** Connecting to a live OMNI or NOAA feed would turn the dashboard from an analytical tool into a real-time situational-awareness platform.
- **Additional panels.** An event spectrogram and a dedicated single-storm inspector, both sketched during development, would deepen single-event analysis.
- **Shareable analysis state.** Letting users save or share the current date range, selected storm, and cursor position would support collaboration and revisiting an insight later.
- **Validated exposure model.** Replacing the heuristic score with a published radiation-belt or particle-flux model would make the exposure simulator quantitatively defensible.
- **Forecast overlay.** Overlaying a short-term forecast alongside the observed record would let operators compare predicted and actual conditions.

10. Consultation Suggestions as Final Results

Presenting ourselves as big data visualization consultants, and based on the analytical surface this dashboard provides, we would offer the following suggestions to a team using it for space-weather monitoring or satellite operations planning:

1. The Threat Escalation Flow should be used routinely to audit which solar wind driver types are currently contributing most to storm-hour outcomes, since this balance can shift across a solar cycle.
2. The Seasonal Pattern panel should be re-checked against each newly loaded date range rather than assumed constant, since the strength of the equinoctial effect is not perfectly uniform across years and solar-cycle phase.

3. Operators of GEO and MEO assets should prioritize the Orbital Exposure Simulator’s danger-threshold alerts during cataloged severe and intense storms, since those shells’ exposure scores are weighted most heavily toward AE index and magnetopause compression, both of which move fastest during shock arrival.
4. The Compare-versus-storm feature in the Time Series panel is well suited to building an informal case library of storms that behaved like a given one, which can support faster qualitative triage during a live event.
5. Because the storm catalog is generated by a simple, transparent rule, it should be treated as a first-pass screening tool, and borderline or short-duration events may warrant manual review directly in the Time Series panel.
6. As more years of OMNI data become available, the pipeline can simply be re-run to extend the dataset, since no part of the dashboard assumes a fixed date range.

11. Conclusion

The Solar Wind and Space Weather Analytics platform stands as a comprehensive and interactive tool for exploring the coupling between the Sun and the Earth. By transforming a large, forbidding, multi-channel record into linked visual narratives, the tool lets users uncover temporal, statistical, categorical, and operational patterns in space weather with ease. From tracking the onset and recovery of a storm, to separating solar wind populations in phase space, to tracing the pathway from driver to storm, to watching the magnetopause compress around a satellite’s orbit, the dashboard serves researchers, forecasters, operators, and students alike.

The system’s modular architecture keeps a clean separation between data preprocessing, the service layer, and dynamic plotting, which makes the application scalable, extensible, and maintainable. Each panel offers a distinct perspective, and the shared toolbar binds them into a single coherent surface. Special attention was given to interactivity, physical grounding, and usability, through custom filters, linked brushing, hover-enabled readouts, and a shared playback cursor.

Ultimately, the tool reinforces the role of visualization in turning a superb but inaccessible public dataset into actionable understanding. By making complex space weather patterns digestible, it supports informed decision-making and lays a foundation for future integration with real-time feeds, validated exposure models, and forecast overlays.

12. Code and Deployment Links

The source code for this project is available in a single GitHub repository.

- **Repository:** https://github.com/cjaitej/BigData_Project

Prerequisites

- Node.js version 18 or higher, with npm.
- Python 3.10 or higher, with pip.

Setup

```
git clone https://github.com/cjaitej/BigData_Project
cd BigData_Project
```

```
npm install
cd server && pip install -r requirements.txt && cd ..
```

Run (two terminals)

```
# Terminal 1 - backend (Flask API on port 5000)
cd server && python app.py

# Terminal 2 - frontend (Vite dev server on port 5173)
npm run dev
```

Open <http://localhost:5173> in a browser once both processes are running.

Technology stack. The data engineering and backend layer uses a Python pipeline (pandas, numpy, requests) to acquire and clean the OMNIWeb HRO data, derive physical quantities, and build a rule-based storm catalog, served through a Flask with Flask-CORS REST API backed by the cleaned CSV. The frontend uses React (with Vite as the build tool) and the D3.js library for all custom visualizations, including a hand-built SVG Sankey diagram and a Canvas-based physics animation, styled with Tailwind CSS, to build an interactive, linked-views dashboard for solar wind and geomagnetic storm exploration.

13. Member Contributions

Each member of Group 21 contributed to the design, implementation, and documentation of this project. Table 3 summarizes the primary contribution of each member.

Member	Contribution
Gaurav Singh Bora (251110031)	Data pipeline and backend (Flask REST API).
Aryan Arora (251110016)	Orbital Exposure Simulator; interactions and frontend.
Jatin Jangra (251110038)	Orbital Exposure Simulator; interactions and frontend.
Ajith Addala (251110006)	Seasonal Pattern panel and frontend.
Choudam Jaitej (251110023)	Threat Escalation Flow; interactions and frontend.
Robin Shah (251110608)	Phase Space Explorer.
Tanmay Saurave (251110618)	Time Series Overview.

Table 3: Contribution of each team member to the project.

Appendix A: Machine Learning — Storm Risk Forecasting

This appendix documents a predictive capability added after the initial five dashboard panels: a *Storm Risk Forecaster* that estimates the probability of a geomagnetic storm occurring in the near future, given the current and recent solar wind state. Where the five panels described in Section 8 are retrospective and exploratory — they help a user understand storms that have already happened — this component is forward-looking, and is exposed in the application as a dedicated *Storm Risk (ML)* tab alongside the main dashboard.

A.1 Motivation

A researcher or operator using the dashboard can identify storms after the fact and study their drivers, but none of the five panels answer the operationally important question: is a storm likely in the next few hours, given what the solar wind is doing right now. This appendix adds that capability as a supervised learning problem layered on top of the same cleaned OMNI dataset and the same storm definition already used by the storm catalog (Section 5.5), so the predictive model and the descriptive panels stay consistent with one another.

A.2 Problem Formulation

For each of three forecast horizons $H \in \{1, 3, 6\}$ hours, a binary classifier predicts whether a storm (SYM-H below -50 nT, the same rule used by `build_storm_catalog()`) occurs at any point in the next H hours, given the solar wind state at the current hour. Three independent classifiers are trained, one per horizon, rather than a single multi-output model, so that each horizon’s decision boundary can be optimized on its own.

A.3 Feature Engineering

Each training row is built from the same processed OMNI table already used by the dashboard, expanded with lag and rolling-window features so a single row carries short-term history rather than just an instantaneous reading:

- **Base solar wind state (8 features):** B_z and B_y (GSM), bulk flow speed, proton density, computed dynamic pressure, IMF magnitude, the AE index, and a binary southward- B_z flag.
- **Lagged history (8 features):** the value of B_z , flow speed, proton density, and dynamic pressure 1 hour and 3 hours prior, since solar wind driving of the magnetosphere is strongly persistent hour-to-hour.
- **Short-window aggregates (3 features):** the 3-hour mean B_z , the number of the last 3 hours with southward B_z , and the 3-hour maximum flow speed — these distinguish a single noisy reading from sustained driving.

- **Current storm state (1 feature):** the current SYM-H value, since an already-active storm is informative about whether it persists into the forecast window.

This gives 20 input features per row, computed identically at training time and at prediction time by a single shared feature-construction function, so a model is always fed the columns it was trained on.

A.4 Model and Training

A `RandomForestClassifier` (300 trees, max depth 12, minimum 5 samples per leaf, class-balanced weighting) is trained independently for each horizon. Random forests were chosen deliberately over a sequence model such as an LSTM: the persistence noted above means a handful of lag and rolling features already carry most of the predictive signal, and a random forest handles both the strong class imbalance (storm hours are roughly 3% of all hours) and the nonlinear thresholds in the data (for example, a southward-Bz cutoff) without extensive tuning, while remaining fast to train and straightforward to validate.

The data is split *chronologically*, with the most recent 15% of the timeline held out as a test set, rather than a random row-wise split. A random split would place hours from the same storm on both sides of the split, letting the model see the "future" of a storm it is also being tested on, which would overstate its accuracy. The chronological split is a stricter, more realistic test: the model is evaluated only on storms it has never seen any part of during training.

A.5 Results

Table 1 reports test-set performance for each horizon.

Horizon	ROC AUC	Precision (at 0.5)	Recall (at 0.5)
+1 hour	0.9970	0.7614	0.9741
+3 hours	0.9915	0.6565	0.9404
+6 hours	0.9783	0.5795	0.8848

Table 1: Storm risk forecaster performance on a chronologically held-out test set, by forecast horizon.

Recall stays high across all three horizons (88–97%), meaning the model rarely misses an actual storm within its forecast window — the property that matters most for an early warning signal, where a missed storm is far costlier than a false alarm. Precision declines as the horizon lengthens (76% at +1h down to 58% at +6h), which is expected: further from the event, more borderline non-storm hours resemble genuine storm precursors, so the model over-predicts risk somewhat more often at longer lead times. All three horizons achieve a ROC AUC above 0.97, indicating strong separation between storm and non-storm hours at every lead time tested.

A.6 Serving the Model

The model is trained once, offline, by `train_storm_model.py`, and the fitted classifiers are serialized to a single `joblib` file together with their feature list and horizons. This file is not checked into version control, both because of its size and because it is fully reproducible from the processed dataset; it is regenerated by re-running the training script. At startup, the Flask service loads this file once and serves three additional endpoints: `/api/model/info` for training metadata and the metrics above, `/api/predict` for a point prediction across all three horizons at a chosen timestamp, and `/api/predict/range` for a risk curve across a date range, which lets the model’s output be plotted directly against the actual cataloged storm intervals for visual validation.

The *Storm Risk (ML)* tab in the frontend surfaces both of these: a user can pick any timestamp in the dataset and see the predicted storm probability at +1h, +3h, and +6h with a risk label (Low, Elevated, High), and can load a risk-over-time curve for a chosen horizon across a date range, with the actual storm hours shaded for comparison — for example, checking that predicted risk climbs ahead of the October 2003 Halloween storms.

A.7 Limitations

The single chronological train/test split is a meaningful improvement over a random split, but it is still only one split; a walk-forward validation across multiple solar cycles would give a more robust estimate of how the model generalizes across different phases of solar activity. As with the orbital exposure simulator (Section 8.5), this component is intended as a decision-support signal to prioritize attention, not as a certified operational forecasting product.